H. PLC Data Type Reference

Appendix H of the EFC Gases & Advanced Materials Controls Standard, Revision H, October 2, 2026. Standalone copy; see the full document for the other chapters and appendices.

This appendix lists the PLC data types (DDTs) used in EFC's Control Expert projects. Section 10.4 of the Controls Standard explains how they are used. The first three tables are the minimum member lists defined by this standard; the remaining tables document the types in the current program library, taken from the Control Expert data editor. Descriptions marked *TBD* have not been confirmed yet.

[Naming in the current library is not fully consistent: `AI_Sensors` has no `DDT_` prefix, and member capitalization differs between types (for example `Auto_Perm` in `DDT_Valve` and `AUTO_PERM` in `DDT_Motor_VFD`). Align these the next time the library is revised.]

H.1 DDT_Process_Value

Table H.1: DDT_Process_Value Minimum Members

Member	Type	Description
Raw	INT or DINT	Unscaled channel value.
Raw_Min / Raw_Max	DINT	Configured raw endpoints.
PV	REAL	Scaled engineering value.
Eng_Min / Eng_Max	REAL	Engineering range endpoints.
Units	STRING[16]	Engineering-unit text or enumerated unit identifier.
Quality	DINT	Valid, uncertain, bad, stale, simulated, or platform equivalent.
Timestamp	DDT or native time	Time of last valid update where required.
LL / L / H / HH	DDT_AI_Alarm	Analog alarm instances.
HW_Fault	BOOL	Module or channel diagnostic fault.
Underrange / Overrange	BOOL	Signal is below/above valid instrument range.
Simulated	BOOL	Value is under authorized simulation.

H.2 DDT_AI_Alarm

Table H.2: DDT_AI_Alarm Minimum Members

Member	Type	Description
SP	REAL	Rationalized alarm setpoint.
DB	REAL	Deadband in engineering units.
OnDelay / OffDelay	TIME	Qualified activation and clearing delays.
Enabled	BOOL	Alarm is configured for use.
Active	BOOL	Qualified alarm condition exists.
Latched	BOOL	Condition has latched where required.
Acked	BOOL	Acknowledgment state when managed in PLC.
Suppressed	BOOL	State-based designed suppression is active.
Shelved	BOOL	HMI/SCADA shelving indication when exchanged.
Priority	INT	Rationalized priority enumeration.

H.3 DDT_Device

Table H.3: DDT_Device Minimum Members

Member	Type	Description
ModeReq / ModeAct	INT or enum	Requested and active operating mode.
StartReq / StopReq / ResetReq	BOOL	Momentary validated requests.
RunCmd	BOOL	Final logical run command before output mapping.
Running / Stopped	BOOL	Processed equipment status.
AutoPerm / ManPerm	BOOL	Mode-specific permissive summaries.
Interlock	DDT_Interlock	Protective/control interlock summary.
FeedbackFault	BOOL	Command and feedback disagree after timeout.
Fault / Warning	BOOL	Aggregated condition summaries.
Available	BOOL	Equipment is healthy and eligible for control.
RuntimeSec	UDINT	Retentive accumulated runtime with controlled reset.
CommOK / DataValid	BOOL	Communications and exchanged data are valid.
Simulated	BOOL	Authorized simulation is active.

H.4 AI_Sensors

Analog sensor data with its four alarm states.

Table H.4: AI_Sensors Members

Member	Type	Description
`PV`	REAL	Scaled process value in engineering units.
`Raw`	REAL	Unscaled value from the input channel.
`Rate`	REAL	Rate of change of the process value *TBD*(units per time).

Table H.4 continued

Member	Type	Description
`Fraction`	REAL	*TBD*
`LL_Alarm`	BOOL	The low-low alarm is active.
`LL_Alarm_Acked`	BOOL	The low-low alarm has been acknowledged.
`Lo_Alarm`	BOOL	The low alarm is active.
`Lo_Alarm_Acked`	BOOL	The low alarm has been acknowledged.
`Hi_Alarm`	BOOL	The high alarm is active.
`Hi_Alarm_Acked`	BOOL	The high alarm has been acknowledged.
`HH_Alarm`	BOOL	The high-high alarm is active.
`HH_Alarm_Acked`	BOOL	The high-high alarm has been acknowledged.

H.5 DDT_Discrete

Discrete input with an override for testing and maintenance.

Table H.5: DDT_Discrete Members

Member	Type	Description
`Override_Enb`	BOOL	Override is enabled for this input; the PLC uses `Override_Sig` instead of the field signal.
`Override_Sig`	BOOL	Value used for the input while the override is enabled.
`Fault`	BOOL	Input fault (for example a channel or wiring fault) *TBD*.

H.6 DDT_Setpoint

Operator setpoint with a default value.

Table H.6: DDT_Setpoint Members

Member	Type	Description
`SP`	REAL	Active setpoint value.
`Default`	REAL	Default value loaded on first scan or reset *TBD*.

H.7 DDT_Valve

On/off valve with limit switch feedback; used with `DFB_Valve`.

Table H.7: DDT_Valve Members

Member	Type	Description
`REQ`	BOOL	Open request from the program (automatic mode).
`Auto_Perm`	BOOL	Permissive summary for automatic operation; all conditions met.

Table H.7 continued

Member	Type	Description
Man_Perm	BOOL	Permissive summary for manual operation; all conditions met.
HMI_Auto	BOOL	SCADA request to switch to Auto.
HMI_Manual	BOOL	SCADA request to switch to Manual.
HMI_Off	BOOL	SCADA request to switch to Off.
HMI_Lockout	BOOL	SCADA request to switch to Lockout.
CMD	BOOL	Final open command to the output.
ZSO	BOOL	Open limit switch feedback (valve is open).
ZSC	BOOL	Closed limit switch feedback (valve is closed).
HMI_CMD	BOOL	SCADA open/close command used in Manual mode.
Interlock	BOOL	Interlock active; the valve is forced to its safe state.
Cycle_Count	DINT	Number of open/close cycles (maintenance counter).
Reset_Count	BOOL	Request to reset the cycle count.
Feedback_Enabled	BOOL	Limit switch feedback is fitted and checked (enables feedback-fault detection).
Auto	BOOL	Valve is in Auto mode.
Manual	BOOL	Valve is in Manual mode.
Off	BOOL	Valve is in Off mode.
LockedOut	BOOL	Valve is locked out.

H.8 DDT_Motor_VFD

Motor driven by a variable frequency drive.

Table H.8: DDT_Motor_VFD Members

Member	Type	Description
REQ	BOOL	Start request from the program (automatic mode).
INTERLOCK	BOOL	Run interlock active; the motor is stopped.
AUTO_PERM	BOOL	Permissive summary for automatic operation.
AUTO	BOOL	Motor is in Auto mode.
MANUAL	BOOL	Motor is in Manual mode.
OFF	BOOL	Motor is in Off mode.
LOCKOUT	BOOL	Motor is in the locked-out (service) mode *TBD*.
HMI_AUTO	BOOL	SCADA command to switch to Auto.
HMI_MANUAL	BOOL	SCADA command to switch to Manual.
HMI_OFF	BOOL	SCADA command to switch to Off.
HMI_LOCKOUT	BOOL	SCADA command to switch to Lockout.
CMD	BOOL	Final motor run command to the drive.
YI	BOOL	Motor running feedback from the drive.
WARNING	BOOL	Motor warning is active.
ALARM	BOOL	Motor alarm (fault) is active.

Table H.8 continued

Member	Type	Description
SPD_AUTO	REAL	Speed command used in Auto mode (from the program).
SPD_MANUAL	REAL	Speed command used in Manual mode (from SCADA).
SPD_CMD	REAL	Output speed command sent to the drive.
RUNTIME	REAL	Accumulated runtime of the motor *TBD*(units).
HMI_RESET_RUNTIME	BOOL	SCADA request to reset the runtime.

H.9 DDT_Scale

Scale or weight transmitter interface.

Table H.9: DDT_Scale Members

Member	Type	Description
GrossWeight	REAL	Gross weight reading.
TareWeight	REAL	Stored tare weight.
NetWeight	REAL	Net weight (gross minus tare).
TareCmd	BOOL	Command to tare the scale.
ZeroCmd	BOOL	Command to zero the scale.
ResetScale	BOOL	Command to reset the scale or its communications *TBD*.
Stable	BOOL	Weight reading is stable (no motion).
Warning	BOOL	Scale warning is active.
Alarm	BOOL	Scale alarm is active.
Fault	BOOL	Scale or communication fault.
Msg	STRING	Status or error message text.

H.10 DDT_Seq

Sequence control and status.

Table H.10: DDT_Seq Members

Member	Type	Description
Disabled	BOOL	Sequence is disabled.
Ready	BOOL	Sequence is ready to start (all start conditions met).
Running	BOOL	Sequence is running.
Msg	string[255]	Current status message for the operator.
Start	BOOL	Start request.
Abort	BOOL	Abort request; the sequence goes to its safe state.
ShowCont	BOOL	*TBD*(likely shows the Continue prompt on SCADA).
Pop_Msg	string[255]	Message shown in an operator pop-up *TBD*.
State	DINT	Current step (state) number.
Continue	BOOL	Operator confirmation to continue past a hold point.

Table H.10 continued

Member	Type	Description
Alarm	DINT	Active sequence alarm code *TBD*.
Complete	BOOL	Sequence has completed.
Percent_Complete	REAL	Progress of the sequence, 0–100%.
HMI_Auto	BOOL	SCADA request to switch to Auto.
HMI_Manual	BOOL	SCADA request to switch to Manual.
HMI_Off	BOOL	SCADA request to switch to Off.
Auto	BOOL	Sequence is in Auto mode.
Manual	BOOL	Sequence is in Manual mode.
Off	BOOL	Sequence is in Off mode.
Ack	BOOL	Operator acknowledgment *TBD*.
HMI_Pause	BOOL	SCADA request to pause the sequence.
Paused	BOOL	Sequence is paused.
HMI_Topfill	BOOL	SCADA request for a top-fill *TBD*.

H.11 DDT_Station

Application-specific type for a cylinder fill station; used with DFB_Station.

Table H.11: DDT_Station Members

Member	Type	Description
TargetWeight	REAL	Target fill weight.
ReadyToStart	BOOL	Station is ready to start *TBD*.
ReadyToFill	BOOL	Station is ready to fill *TBD*.
Active	BOOL	Station is filling.
Complete	BOOL	Station has successfully completed the fill *TBD*.
Deadband	REAL	Fill deadband around the target weight *TBD*.
HMI_Select	BOOL	SCADA request to select this station.
HMI_Reset	BOOL	SCADA request to reset the station.
Selected	BOOL	Station is selected.
Venting	BOOL	Station is venting.
Vacuuming	BOOL	Station is being evacuated.
Recovering	BOOL	Station is recovering gas *TBD*.
Vacuumed	BOOL	Evacuation is complete.
Press_Balanced	BOOL	Pressures are balanced *TBD*.
Disabled	BOOL	Station is disabled.
Failed	BOOL	Station fill has failed.
PT_Valid	BOOL	Station pressure transmitter reading is valid.
PT_Diff	REAL	Pressure difference *TBD*.
InletREQ	BOOL	Inlet valve request.
InletCMD	BOOL	Inlet valve command.
SampleREQ	BOOL	Sample valve request.

Table H.11 continued

Member	Type	Description
SampleCMD	BOOL	Sample valve command.
VacuumREQ	BOOL	Vacuum valve request.
VacuumCMD	BOOL	Vacuum valve command.
PT	REAL	Station pressure.
Press_Max	REAL	Maximum allowed station pressure *TBD*.
TareCMD	BOOL	Command to tare the station scale.
ZeroCMD	BOOL	Command to zero the station scale.
GrossWeight	REAL	Gross weight.
TareWeight	REAL	Tare weight.
NetWeight	REAL	Net weight.
StatusEnum	INT	Station status code (see the StatusEnum values in Appendix I of the Controls Standard).
Msg	string[255]	Status message for the operator.
ScaleLoaded	BOOL	A cylinder is on the scale *TBD*.
ScaleTared	BOOL	Scale has been tared.
Fill_Rate	REAL	Fill rate *TBD*(units).
HMI_SP_Max_Vac_Press	REAL	Operator setpoint: maximum pressure at which the vacuum lines may be opened (permissive setpoint, Section 5.6 of the Controls Standard).
TotalWeight	REAL	Total weight *TBD*.
SampleWeight	REAL	Sample weight *TBD*.
SamplePress	REAL	Sample pressure *TBD*.
Pressurizing	BOOL	Station is building pressure.
Sampling	BOOL	Station is sampling.
HighPressWarningReq	BOOL	Station maximum-pressure warning request *TBD*.
PressurizeReqd	BOOL	Pressurizing is required *TBD*.
SampleReqd	BOOL	Sampling is required *TBD*.
Filling	BOOL	Station is filling.
Disconnecting	BOOL	Station is disconnecting.
HMI_Disc	BOOL	SCADA request to disconnect.
Purged	BOOL	Station has been purged.
Light	BOOL	Station indicator light *TBD*.